Depth Estimation with Single-camera Motion Stereo in Visual-aid Odometry

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Abstract

Visual-aid odometry is wildly used for reducing the accumulate error of ordinary odometry due to drifted motion sensor measure over time. Depth estimation is a key procedure in visual-aid odometry. Traditional depth measurement techniques require multiple camera perspectives, which is costly. In pursuit of low-cost, the presented work proposed a single-camera depth estimator with motion stereo. A MATLAB visual-aid odometry demonstrates the proposed estimator can reduce the accumulate error of ordinary odometry significantly.

1 Introduction

Odometry refers to the use of data from motion sensors to estimate change in position over time. It is wildly used to estimate position of a robot or unmanned vehicle relative to a starting location, as stated by (1).

$$P_{estimate}(t) = P_{start} + \int V_{measured}(t)dt \tag{1}$$

However, in practice, motion sensors are very likely to drift over time [2]. So the velocity is measured with error and the integration in (1) accumulates the error, causing inaccurate estimation. Therefore, adding extra sensors to calibrate motion sensors measurement becomes a common practice in this field, in which the introduction of vision sensor like camera is called visual-aid odometry.

In visual-aid odometry, a common operation is to use object's coordinates in image frame to solve for its corresponding coordinates in world frame, which requires the depth of the object relative to the camera in addition to camera intrinsic and extrinsic parameters. Therefore, the depth estimation problem arises in the context of visual-aid odometry.

Traditional depth measurement techniques require multiple camera perspectives and calculate position of target by stereo triangulation, as shown in Figure 1a. However, a drawback of multi-camera system is it's relatively costly no matter in calibration consensus or multi-source data time-stamp alignment.

Motivate by this situation, the presented work proposed a depth estimation techniques for moving target with only one camera in pursuit of low-cost. The principle we utilized is the kinematic equivalence of moving-object and moving-camera. As shown in Figure 1b, by using the measurement from two consecutive video frames, we applied a motion stereo triangulation to substitute the traditional depth estimation triangulation. The detail and effectiveness of the proposed depth estimator is demonstrated in the following sections in this paper.

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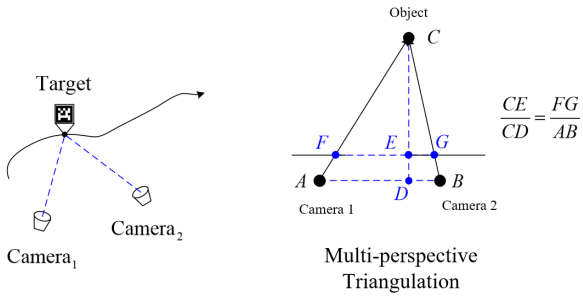
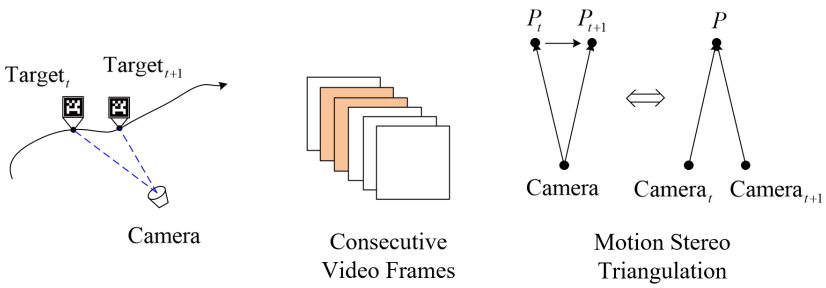


(a) Traditional multi-perspective triangulation.



(b) The proposed motion stereo triangulation.

Figure 1: Depth estimation. (a) is the traditional multi-perspective triangulation. (b) is the proposed motion stereo triangulation.

1.1 Related Work

Papers about learning-free single-camera depth estimation are quite few. Rosenfeld [1] proposed the technique that use one moving camera to estimate two static static camera with different viewing angle in depth measurement, which directly motivates our proposed project.

2 The Proposed Algorithm

2.1 Problem Statement

The visual-aid odometry problem we discussed in this paper can be defined rigorously as follows. Given the target's starting position ${}^{W}P_0$ in world frame, a camera with know intrinsic parameter K and extrinsic parameter $\begin{bmatrix} {}^{W}R_C & {}^{W}t_C \\ 0_{1\times3} & 1 \end{bmatrix}$, and a motion sensor measures the target's real-time velocity (with error) ${}^{W}V(t)$ respect to world frame, we aims to calculate target's real-time position ${}^{W}P_t$ in world frame. An illustration is demonstrated in Figure 2.

2.2 Pipeline

To perform the defined task, a pipeline is proposed as shown in Figure 3. The pipeline adopts an iterative manner: next estimation ${}^{W}P_{t+1}$ is calculated based on previous estimation ${}^{W}P_t$. Hence, given the initial position of the target ${}^{W}P_0$, the pipeline is capable to calculate subsequent positions by iteration.

At first, the proposed method transform all measurements in world frame at time stamp t to camera frame with camera extrinsic parameters.

$$\begin{bmatrix} {}^{C}P_t \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{W}R_C & {}^{W}t_C \\ 0_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} {}^{W}P_t \\ 1 \end{bmatrix} \tag{2}$$

$${}^{C}V_t = {}^{W}R_C {}^{W}V_t \tag{3}$$

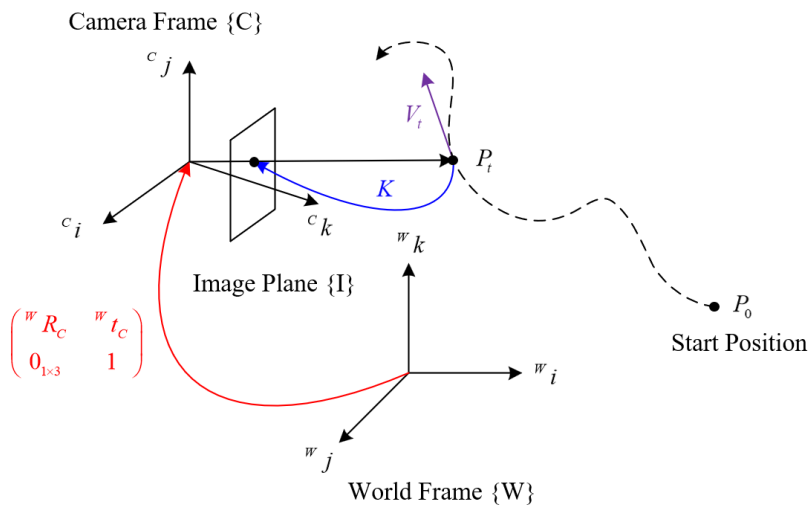


Figure 2: Illustration of problem statement

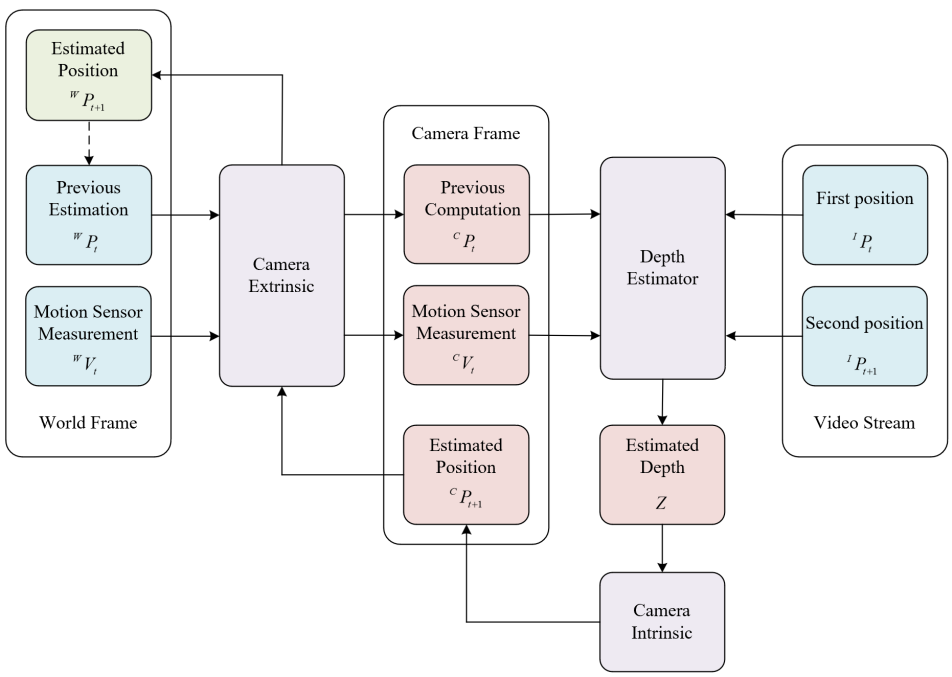
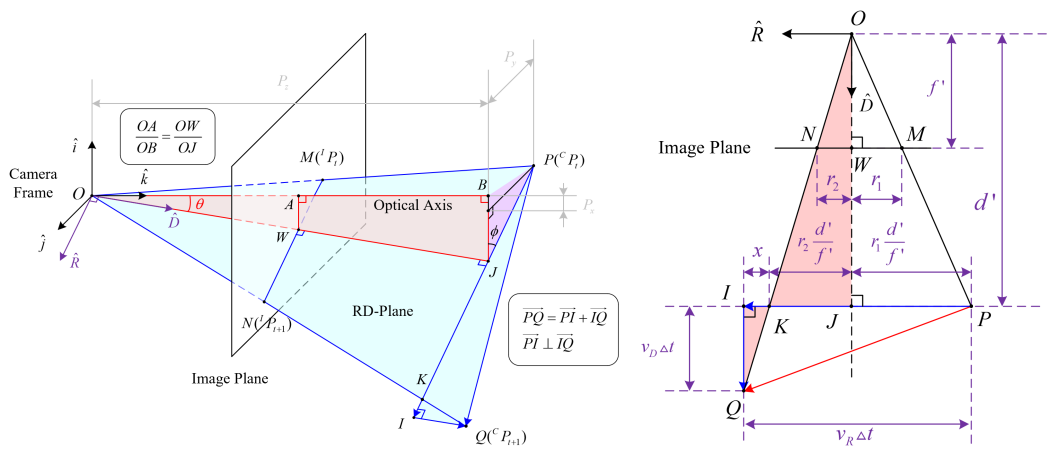


Figure 3: Pipeline of the visual-aid odeometry with proposed single camera depth estimator. The blue blocks denote input variable for one estimation loop, and the green block denotes the output variable.

Then, with the measurement ${}^{C}P_t, {}^{C}V_t$ in camera frame and target's image coordinates with consecutive time stamps ${}^{I}P_t, {}^{I}P_{t+1}$, a depth estimator will estimate thge target's depth at time stamp $t+1$.

Once the depth is acquired, the target's position in world frame at time stamp $t+1$ can be calculated via inverse operation of camera projection, which can be down with inverse camera intrinsic and extrinsic parameters.

The depth estimator and inverse camera projection will be introduced in detail in the following subsections.



(a) Localization of RD-plane

(b) Motion stereo triangulation in RD-plane

Figure 4: Depth estimation. (a) is the traditional multi-perspective triangulation. (b) is the proposed motion stereo triangulation.

2.3 Depth Estimation

In the presented work, depth estimation is performed with geometry shown in Figure 4. We consider the plane spanned by two vector $\{\overrightarrow{{}^{C}P_t}, \overrightarrow{{}^{C}P_{t+1}}\}$ in camera frame, which is named by RD-plane, as shown in the blue plane in Figure 4a. For simplification, we rename these two point ${}^{C}P_t, {}^{C}P_{t+1}$ by P, Q, respectively. Denote the corresponding points of P, Q in image plane by M, N, respectively.

To determine a set of basis of RD-plane, we decompose the displacement vector $\overrightarrow{PQ}$ into two components

$$\overrightarrow{PQ} = \overrightarrow{PI} + \overrightarrow{IQ} \tag{4}$$

where $\overrightarrow{PI} \parallel \overrightarrow{MN}$ and $\overrightarrow{IQ} \perp \overrightarrow{MN}$. Hence a basis $\{\hat{R}, \hat{D}\}$ for RD-plane is obtained. Both of $\{\hat{R}, \hat{D}\}$ originates at the camera optical center, $\hat{R}$ points along $\overrightarrow{PI}$ and $\hat{D}$ points along $\overrightarrow{IQ}$.

As shown in Figure 4a, the RD-plane is localized with two angles $< \hat{j}, \hat{R} >$ and $< \hat{k}, \hat{D} >$. Given ${}^{C}V_t = \begin{bmatrix} {}^{C}V_{tx} & {}^{C}V_{ty} & {}^{C}V_{tz} \end{bmatrix}^\top$ and ${}^{C}P_t = \begin{bmatrix} {}^{C}P_{tx} & {}^{C}P_{ty} & {}^{C}P_{tz} \end{bmatrix}^\top$, The former can be computed through $\angle BJP$, denotes it by ϕ.

$$\phi = \arctan \frac{|{}^{C}V_{ty}|}{|{}^{C}V_{tx}|} \tag{5}$$

The later can be computed through $\angle BOJ$, denotes it by θ

$$\theta = \arctan \frac{|{}^{C}P_{tx}||{}^{C}V_{tx}| + |{}^{C}P_{ty}||{}^{C}V_{ty}|}{|{}^{C}P_{tz}||{}^{C}V_{tx}|} \tag{6}$$

Next let's focus on $\triangle OBJ$ locates in the normal plane of the image plane (the red plane in Figure 4a), we have

$$|OW| = \frac{|OA|}{\cos\theta} = \frac{\text{focal length}}{\cos\theta} \tag{7}$$

$$|OJ| = \frac{|OB|}{\cos\theta} = \frac{\text{depth}}{\cos\theta} \tag{8}$$

which are two of the key equations used to estimate depth. To compute the value of $|OW|$ and $|OJ|$, we applied an triangulation in RD-plane, as shown in Figure 4b. In the RD-plane, we have

$\triangle OKJ \sim \triangle QKI$ (two red triangles highlighted in Figure 4b), then we have

$$\begin{aligned} \frac{|IK|}{|KJ|} &= \frac{|IQ|}{|OJ|} \\ \Rightarrow \frac{|PI| - |PJ| - |KJ|}{|KJ|} &= \frac{|IQ|}{|OJ|} \end{aligned} \tag{9}$$

where

$$|PI| = \frac{\sqrt{|{}^C V_{tx}|^2 + |{}^C V_{ty}|^2}}{\sin\phi} \triangle t \tag{10}$$

$$|IQ| = \frac{|{}^C V_{tz}|}{\sin\phi} \triangle t \tag{11}$$

where $\triangle t$ is the time duration between two consecutive time stamps t and $t+1$.

$$|PJ| = |WM| \frac{|OJ|}{|OW|} \tag{12}$$

$$|KJ| = |WN| \frac{|OJ|}{|OW|} \tag{13}$$

$$|WM| = \sqrt{(u_t)^2 + (v_t)^2} \tag{14}$$

$$|WN| = \sqrt{(u_{t+1})^2 + (v_{t+1})^2} \tag{15}$$

where ${}^I P_t = [u_t \quad v_t \quad 1]^\top$ and ${}^I P_{t+1} = [u_{t+1} \quad v_{t+1} \quad 1]^\top$ are image coordinates of M, N, respectively.

From (5) to (15), we can solve for the estimated depth for P and Q

$$D_P = |OJ| \cos\theta = \frac{|PI||OW| - |IQ||WN|}{|WM| + |WN|} \cos\theta \tag{16}$$

$$D_Q = D_P + {}^C V_{tz} \triangle t \tag{17}$$

D_Q will be used to estimate ${}^W P_{t+1}$.

2.4 Inverse Camera Projection

Given object with coordinates respect to camera frame $[{}^C X, {}^C Y, {}^C Z]^\top$, its corresponding point in image plane $[u, v, 1]^\top$ is

$$\frac{1}{w} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} {}^C X \\ {}^C Y \\ {}^C Z \end{bmatrix} \tag{18}$$

where K is camera's intrinsic matrix. And the inverse of (18) is

$$\begin{bmatrix} {}^C X \\ {}^C Y \\ {}^C Z \end{bmatrix} = {}^C Z \times K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{19}$$

where ${}^C Z$ is the depth of the object respect to camera frame. In this project we set ${}^C Z = D_Q$, $u = u_{t+1}$, and $v = v_{t+1}$. Then

$${}^C P_{t+1} = \begin{bmatrix} {}^C P_{t+1,x} \\ {}^C P_{t+1,y} \\ {}^C P_{t+1,z} \end{bmatrix} = D_Q \times K^{-1} \begin{bmatrix} u_{t+1} \\ v_{t+1} \\ 1 \end{bmatrix} \tag{20}$$

And

$$\begin{bmatrix} {}^W P_{t+1} \\ 1 \end{bmatrix} = \begin{bmatrix} {}^W R_C & {}^W t_C \\ 0_{1\times 3} & 1 \end{bmatrix}^\top \begin{bmatrix} {}^C P_{t+1} \\ 1 \end{bmatrix} \tag{21}$$

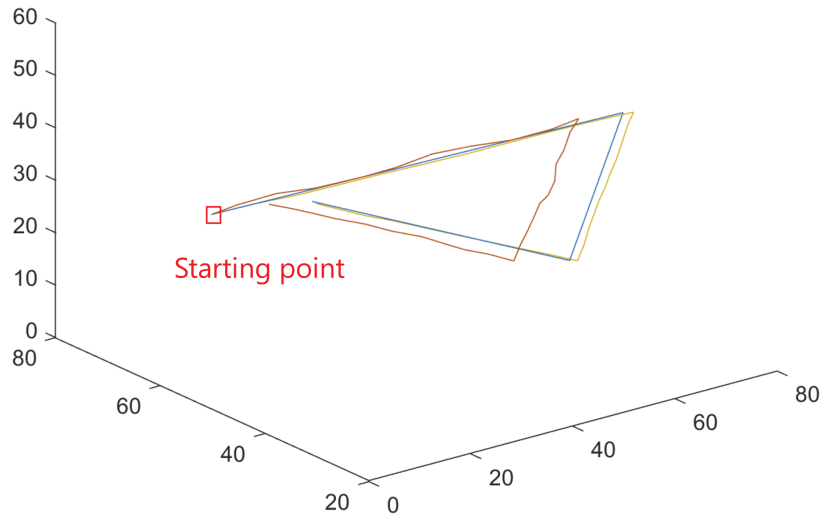
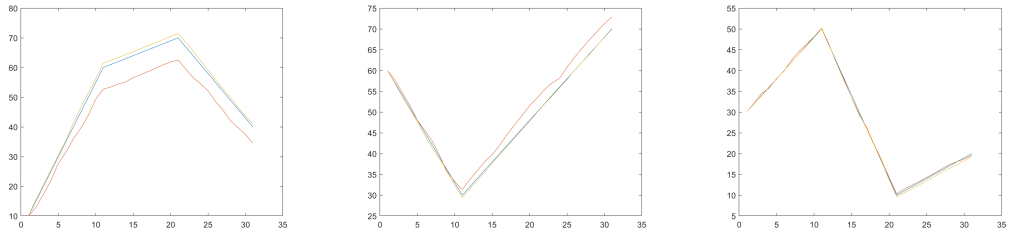


Figure 5: Trajectory plotted from Simulation. Blue: Ground truth. Orange: Odometry. Yellow: Visual-aid odometry.



(a) Component X (b) Component Y (c) Component Z

Figure 6: Components of trajectory plotted from Simulation. Blue: Ground truth. Orange: Odometry. Yellow: Visual-aid odometry.

3 Experiments

To verify the proposed pipeline, a MATLAB simulation is performed. In simulation, the camera intrinsic and extrinsic parameters, target starting position and ground truth of target velocity are all manually set. The motion sensor measurement simulated by ground truth + Gaussian noise (20% magnitude).

To specify, in simulation, the camera parameter used are

$$\begin{bmatrix} {}^{W}R_C & {}^{W}t_C \\ 0_{1\times 3} & 1 \end{bmatrix} = diag\{1,1,1,1\}, \quad K = \begin{bmatrix} 53.3333 & 0 & 20 \\ 0 & 40 & 30 \\ 0 & 0 & 1 \end{bmatrix} \tag{22}$$

A qualitative result of the simulation is shown in Figure 5 and 6. The estimation of ordinary odometry drifts over time while the visual-aid odometry can make the estimation close to the ground truth. However, the numerical results vary widely in the simulation, so a stable quantitative result is not given here. But we can conclude that the error of the visual-aid odometry is smaller than that of the normal odometry.

References

[1] R. Nevatia. Depth measurement by motion stereo. *Computer Graphics and Image Processing*, 5(2):203–214, 1976.

[2] D. Nistér, O. Naroditsky, and J. Bergen. Visual odometry. In *Proceedings of the 2004 IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2004. CVPR 2004.*, volume 1, pages I–I. Ieee, 2004.

Appendix

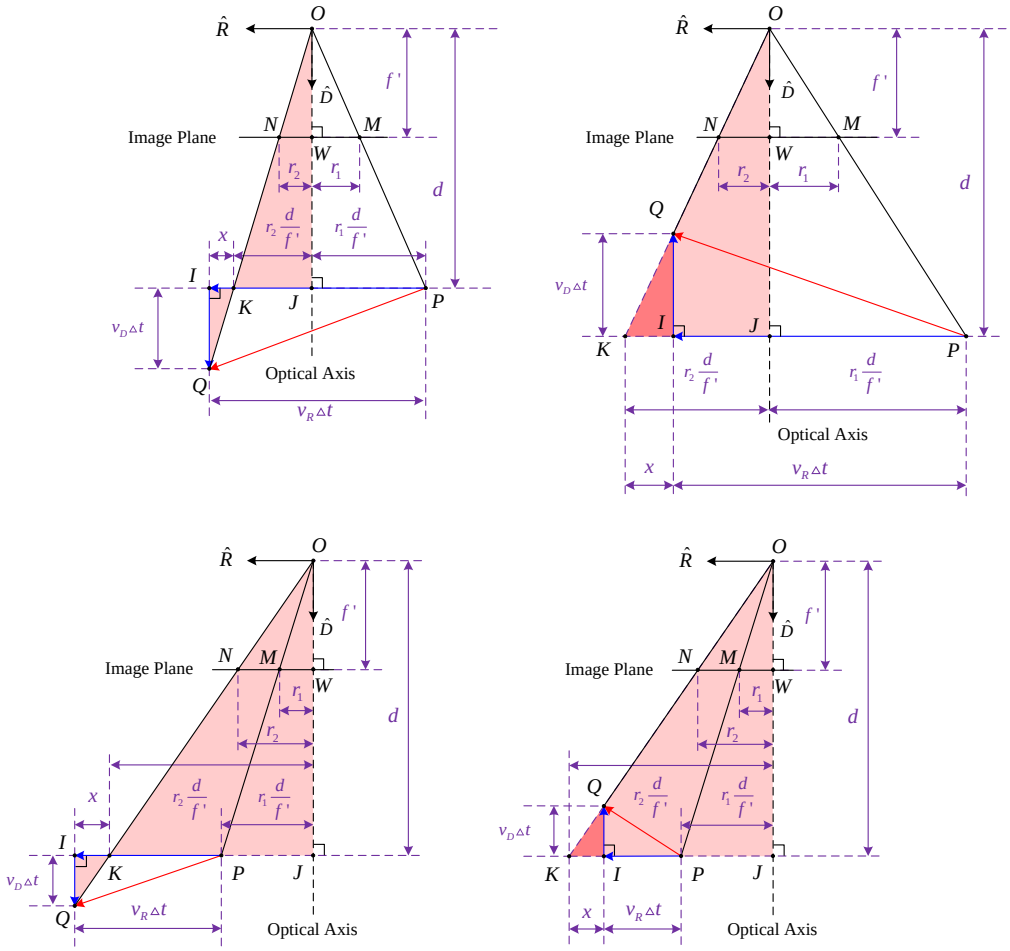


Figure 7: Four possible situations of triangulation in RD-plane. All of them can derive the same result as shown in (16) and (17).